

Day 7-9 – Assignment – ASP.NET Core Web API

ABC Retail wants to build a small Product Management API. Initially they don't want any database. All products should be stored inside memory. When the application restarts, data will be recreated.

Tasks:

1. Create Product Model.
2. Create DTOs
 - a. CreateProductDto
 - b. UpdateProductDto
3. Apply Validations
 - a. Name: Required, Max Length 50
 - b. Category: Required
 - c. Price: Range 1-500000
 - d. Stock: Range 0 to 10000
4. Create Static Product List
 - a. Create ProductStore.cs Use List<Product> with 10 sample records.
5. Create Service Layer
 - a. IProductService and ProductService
 - b. implement
 - i. GetAll()
 - ii. GetById()
 - iii. Add()
 - iv. Update()
 - v. Delete()
 - vi. SearchByCategory()
 - c. Register Service
6. Create Controller: ProductsController
7. Route: api/products
8. Return Proper Status Codes
 - a. GET: 200 OK
 - b. POST: 201 Created
 - c. DELETE: 204 No Content
 - d. Not Found: 404
 - e. Validation Error: 400
9. Sample Data

Id	Name	Category	Price	Stock	Manufacturer	Available
1	Laptop	Electronics	65000	15	Dell	Yes
2	Mouse	Electronics	700	50	Logitech	Yes
3	Keyboard	Electronics	1500	30	HP	Yes
4	Office Chair	Furniture	8500	12	Nilkamal	Yes
5	Study Table	Furniture	12000	6	Godrej	Yes
6	Water Bottle	Accessories	450	100	Milton	Yes
7	Notebook	Stationery	120	300	Classmate	Yes
8	Pen Pack	Stationery	90	500	Cello	Yes
9	Monitor	Electronics	14500	20	Samsung	Yes
10	Printer	Electronics	9800	8	Canon	No



Microsoft DotNet

Free, Open-Source, Cross-Platform Framework



Expected Endpoints

- MethodEndpoint
- GET /api/products
- GET /api/products/{id}
- GET /api/products/category/{category}
- GET /api/products/search?name=laptop
- POST /api/products
- PUT /api/products/{id}
- DELETE /api/products/{id}
- GET /api/products/applicationinfo

☞*☞ Debug your doubts, deploy your success ☞*☞